1. What are ICT.

→ ICT Stand for Information and communication Technology.ICT refers to any device, component, or system that allows people to store, process, retrieve, transmit or manipulate information electronically.

2. **Information Technology (IT)** : The computer hardware (CPU, RAM,HDD,…), Software (OS, apps,game,…) and network (routers, switches) that process and manage data.
3. **Communication Technology (CT) :** Exchange data such as the internet, wifi, bluetooth and satellite systems.
4. **Why are they used in the workplace ?**

→ICT is used in the workplace primarily to boost efficiency, enable communication, and support decision-making. Some reasons ICT is used in the workplace are :

(a)Faster Communication: Email, Instant messaging, vido conferencing.

(b)Automation of repetitive tasks : Software handles payroll, inventory tracking, data entry and report generation.

(c)Data Storage & access : Cloud service, enabling remote work, accessed from any device.

(d)Better decision-making : By Spreadsheets (Excel), database and many tools data quickly providing.

(e)Cost reduction : Digital forms replace paper, video call replace travel….etc all saving time and money.

5. How is ICT helpful in education, e-governance, health care, Business, Entertainment and daily life.

→**Education :** ICT helps students and teachets is online classes & Exams, digital learning, smart boards and fast access to study materials.

E-Government : ICT helps the government provide online services such as bill payment, online forms, digital documents and public information.

Health care : ICT helps doctors and hospitals maintain patient records, online appointments, medical reports, and telemedicine services.

Business : ICT helps businesses in communication, online marketing, data management, online shopping and faster work processing.

Entertainment : ICT provides movies, music, games, social media and online streaming services for entertainment.

Daily Life : ICT helps people in Mobile banking, online shopping, GPS navigation, communication and accessing information quickly.

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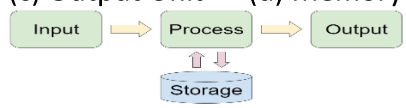
6. Basic Units / Main Components of a computer

→Four main components of a computer

(a) Input Unit (b) Processing Unit (CPU)
(c) Output Unit (d) Memory Unit



IPO Cycle

7.What is Hardware ?

Hardware is the physical part (all peripheral devices) of a computer you can touch.

Example : Keyboard, CPU, Printer, Monitor,…..etc

8. What is software ?

Software is a set of program (collection of instruction) that tells the hardware what to do.

Example: O.S, Word, Photoshop, VLC, ……

9.What is humanware or Peopleware or liveware ?

Humanware refers to the human user who interacts with the computer system.

10.What is CPU ?

CPU is the part of the computer in which arithmetical, logical and control I/O operations and instructions are decoded and than executed.

- Brain of Computer – CPU
- CPU is also called – Microprocessor
- Speed of CPU Measured in – GHz (Giga hertz)
- CPU understand which language – Machine Or Binary language(0,1)
- 1^{st} Microprocessor – Intel 4004 in 1971.

11. How many component / Main feature of CPU ?

The CPU has three main internal components.

(a)ALU (Airthmetic logic Unit)

(b) CU (Control Unit)

(c) MU (Memory Unit) / Registers

12.What is ALU ?

ALU performs all mathematical calculations and logical comarisons.

- Airthmetic stand for : Add, Sub, Mult, Divi
- Logic stand for : Comparisons (<,>,=,<=,>=,!=,≠,…)
- Bitwise operations : AND, OR, NOT, XOR

13. What is CU ?

CU only manage/ Control the all parts of the computer data and instructions. It does not process or store data. It is a part of hardware.

14. What is Register ?

Register is a part of cpu memory. It is extremely small, ultra-fast storage locations inside the cpu. Register store data is temporarily.

15.Which types of component/device are in the input Unit ?
Keyboard, Mouse, Joystic, Track Ball, Light pen, Digitizer, Microphone, Web Camera, Scanner, Biometric Sensor,……

16. Which types of component/device are in the output Unit ?
Monitor, Plotter, Printer, Speaker, Projector.

17. What is I/O device ?
A device that can perform both input and output operations is called an Input output (I/O) device. Example :Touchscreen, Concept keyboard, modem, network card (NIC).

18. Which types of component/device are in the processing unit ?
CPU, Mother board

19. Which types of componenet/ device are in the memory unit ?
Register, RAM, ROM, Cache, HDD, SDD, Pendrive

20. Input Unit Key points
- Input device converts input data into machine-readable (binary / Machine) form.
- Scanner convert hard copy into soft/digital copy.
- Printer convert soft copy into hard copy.
- Mouse,Trackball,Joystick, Light pen, Touch screen is a pointing input device.
- No of key in General purpose keyword is 104-105
- No of button in General purpose mouse is 3.
- Joystick use -Video game,Submarines,Simulations
- Digitizer used by architects & engineers.
- Microphone convert sound waves into audio signal / Electrical Energy.
- OMR detects marked responses on forms.
- Bank Cheques are processed using a MICR scanner.
- In shopping malls, etc,, BCR are used.
- Input device are also known as peripheral devices.

21. Output Unit Keypoints
- Output unit present processed data to the user.
- It convert machine readable result into human readable form
- CRT display produced image when electron beam strikes a phosphor surface.
- LED monitor is an advanced form of LCD monitor.
- Plotter produce hard copies of large graphs and designs on paper.
- Plotter are two part –(i) Drum plotter (ii) flatbed
- Two types of printer (i) impact (ii) Non- impact
- Dot matrix, daisy wheel, drum printer are example of impact printer.
- Laser, inkjet,thermal printer are example of non-impact printer.
- Inkjet printer use ink jet technology and it's consists of a head, ink cartridge, belt,…..etc
- Leaser printer consists laser beam, multi-side mirror, photoconductive drum & toner(powder).
- Thermal printer is uses heat to print letters on paper, No need any ink/toner.

22. Storage Unit key points.
- Main memory is also called primary memory / Semiconductor memory / Internal memory.
- Secondary memory is also called Auxilary memory / External memory / Backing store memory.
- Example of Primary memory is : RAM, ROM, Cache
- Ram is a volatile memory because data is lost when power is off and not store data permanently.
- ROM is a non-volatile memory.
- 3 types of ROM (I) PROM (2) EPROM (3) EEPROM
- 2 Types of RAM (i) SRAM (ii) DRAM
- SRAM used for CPU Cache memory.
- SRAM use flipflop, it does not need refresh.
- DRAM use capacitor, it need refreshing.
- IN EPROM data erase using UV(Ultra violet) light.
- IN EEPROM data erase using electrically.
- Example of Secondary memory is : Floppy disk, hard disk, CD, DVD, Pen drive, SSD…..etc
- Example of magnetic disk : Floppy disk, Hard disk
- Example of Optical Disk : CD, DVD, Blue ray
- Example of Flash memory : SSD, pen drive.
- Floppy disk was first created in 1967 by IBM.
- Hard disk platters made of non-magnetic material usually aluminium alloy and coated magnetic material. Its speed measure in RPM.
- RAM memory is lost when the system is turned off.
- HDD device may be damaged by frequent improper shutdowns ?

23. Two Types of Software : (i) System Software (ii) Application Software

24. System Software : Designed to run a computer hardware and application programs.

25. Application Software :Designed to specific purpose used by end users.

26. System Software : Three types (i) O.S (ii) Utility Software (iii) Language processors.

27. What is Operating System (OS) ?

It is type of system software. It is interface between a computer user and computer hardware.

28. Computer OS name : Microsoft windows, Linux, Ubuntu, macOS, UNIX, Fedora, Debian, ChromeOS.

29. Mobile OS name : Android, iOS, Symbain, Blackberry OS, Windows Phone.

30. What is Utility Software ?

It is system software that helps maintain, manage, protet, and optimize a computer system.

31. Example of Utility Software : Antivirus, Disk cleanup, disk defragmenter, backup software, file compression software, firewall, system restore,..

32. What is Language Processor

It convert source code written in a programming language into machine language / Code.

33. Three types of language processor.

(i) Compiler (ii) Interpreter (iii) Assembler

34. Compiler : Translate and executes the whole program at a time. Example : GCC, Turbo C++

35.Interpreter : Translate and executes one line at a time. Example : Python

36. Example of Application Software :

(a) Word processing : MS word, google docs
(b) Spreadsheet : MS Excel, Google sheet
(c)Presentation : MS powerpoint, Google sheet
(d)Database : MS access, oracle, MYSQL
(e)Graphics : Photoshop, coreldraw, paint
(f)Multimedia : VLC player, Window media player
(g)Communication : Zoom, Gmail, Whatsapp
(h) Accounting : Tally prime
(i)Reservation : IRCTC, Red bus
(j) Food delivery : swiggy, Zomato
(k) e-commerce : amazon, flipkart, meesho,blinkit

37. Mother Board also known as mainboard or System board.

38. Mother board holds all components of the system such as CPU, Main memory, Ports, Peripheral devices and many electronic componets (capacitor, resistor, transistor, diode, fuse, chip,CMOS…..)

39. CPU is a brain of computer system then mother board called is spine/backbone of computer system.

40. Symbol & it's Name

~	Tilde	!	Exclamation mark
@	At sign	#	Hash
^	Caret	&	Ampersand
*	Asterisk	()	Parenthesis
-	Hyphen / Dash	_	Underscore
{}	Curly brace	\	Backslash
:	Colon	;	Semicolon
"	Double quotation	'	Single quotation
<	Less than	>	Greater than

41. Full Form

ALU	Airthmetical logical unit
CU	Control unit
ATM	Automated teller machine
OMR	Optical Mark reader
OCR	Optical Character reader
MICR	Magnetic ink character recognition
BCR	Bar code reader
Bit	Binary digit (0,1)
CRT	Cathod ray tube
LCD	Liquid crystal display
LED	Light emitting diode
DPI	Dot per inch
RGB	Red green blue
CMYK	Cyan magenta yellow black
PPM	Page per minute
RPM	Round per minute
RAM	Random access memory
ROM	Read only memory
SRAM	Static RAM , DRAM → Dynamic RAM
DDR4	Double data rate 4 RAM
PROM	Programmable ROM
EPROM	Erasable Programmable ROM
EEPROM	Electronically Erasable PROM
HDD	Hard disk drive
SSD	Solid state drive
CD	Compact Disc
DVD	Digital Versatile/Video Disc
OS	Operating System
CUI	Command user interface
GUI	Graphical user interface
BIOS	Basic input output system
CMOS	Complementary metal oxide semiconductor
SATA	Serial advanced technology attachment
GPU	Graphics processing Unit
USB	Universal serial bus
LAN	Local area network
VGA	Video graphics array
HDMI	High definition multimedia interface
SMPS	Switched mode power supply
UPS	Un-interruptible power supply
PCI	Peripheral component interconnect
PCIe	PCI express
IC	Integrated Circuit
PCB	Printed Circuit Board

Website visit :
https://bihar24y7.github.io/computer/